

— RAPID REPORT / 2018 – 2022

Modelling House Prices in Cambridgeshire

A rapid statistical study of residential sale prices and the models that predict them.

PREPARED BY

Diana • Matt • Davies • Natsani • Xiangjun • Vedanta



Complexity & Concerns

5 yrs

2018 – 2022

24

property types

VARIABLES IN PLAY

Room profile

Bedrooms · Bathrooms · Reception rooms

Location

MSOA & LSOA

Social profile

Ethnicity · Age · Education · Distance to work

CONCERN → SOLUTION

Dimension inconsistency

→ handled in preprocessing

Bedroom / bathroom counts

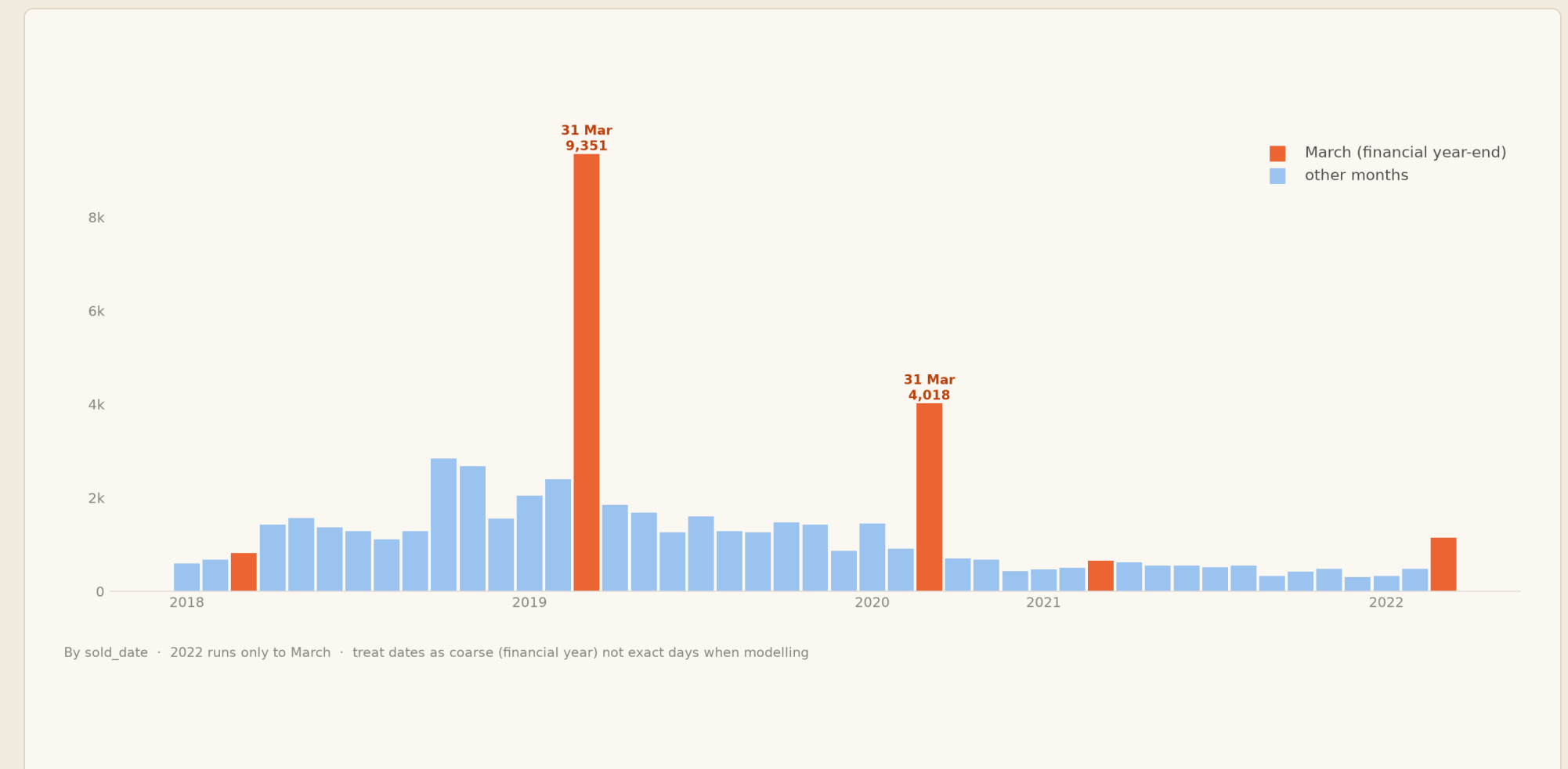
→ resolved in implementation

Date of sale is not exact

→ framed as prediction, not forecast

When sales were recorded

Monthly sales counts, 2018 – 2022



The 31-March artifact: financial year-end dates are imputed onto thousands of records, creating false monthly spikes — treat as coarse dates, not real activity.

Three Modelling Decisions

01 – PHILOSOPHY

Intuitive

vs

Optimised

02 – FEATURE SELECTION

Stepwise

vs

Recursive Feature Elimination

03 – MODEL FRAMEWORK

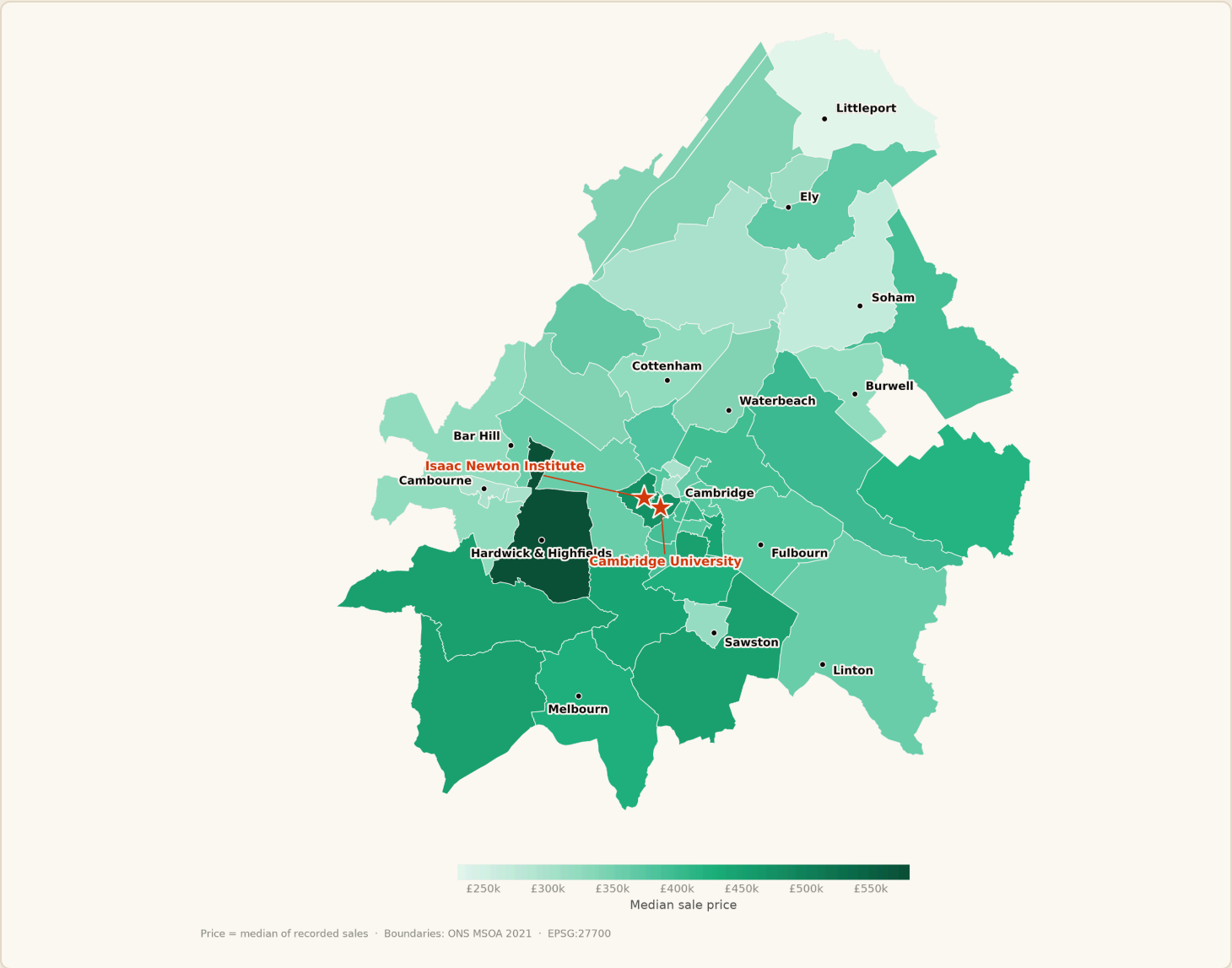
Baseline Linear

vs

Complex Non-linear

House prices across Cambridgeshire

Median sale price by area, 2018 – 2022



Priciest in Cambridge and the villages west of the city.